Name :- DINKY ASWANI
Class :- T11 RollNo :- 2401006
Subject:- ASD&D
Branch :- AI & DS

Aim

To containerize an application using Docker

LO 04

To automate provisioning configuration , container orchestration and cloud resource management using Ansible , Docker , kubernetes and Terraform

Theory

1. What is Docker?

Docker is an open-source platform used to develop, package, and run applications in containers. It bundles an application with all its dependencies, ensuring it works the same on any system. Docker is lightweight, fast, and simplifies application deployment. It is widely used in software development, testing, and cloud environments.

2. What is Containerization?

Containerization is a virtualization technology that packages an application along with its dependencies, libraries, and configuration files into a container. Containers run consistently across different environments while sharing the host operating system kernel. They are lightweight, portable, and start quickly, making application deployment and management easier.

3. Difference Between Docker Image and Docker Container

Docker Image

- Read-only template or blueprint.
- Contains application and dependencies.
- Used to create containers.
- Cannot run by itself.

Docker Container

- Running instance of a Docker Image.

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- Executes the application.
- Can be started, stopped, or deleted.
- Writable while running.

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```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Lenovo>docker run - it
docker: invalid reference format.
See 'docker run --help'.

C:\Users\Lenovo>docker run - it ubuntu /bin/bash
docker: invalid reference format.
See 'docker run --help'.

C:\Users\Lenovo>docker run -it ubuntu /bin/bash
root@6c669b3afecc:/#
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Lenovo>docker run - it
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See 'docker run --help'.

C:\Users\Lenovo>docker run - it ubuntu /bin/bash
docker: invalid reference format.
See 'docker run --help'.

C:\Users\Lenovo>docker run -it ubuntu /bin/bash
root@6c669b3afecc:/#
```

```
C:\Users\Lenovo>docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
ubuntu        latest    de7345b16e94   6 days ago      100MB
my-web-app    latest    9e5a44f3e4d4   3 months ago    1.09GB
<none>        <none>    e02d55cb5e5b   3 months ago    1.09GB
navin_img     latest    73f549b84804   3 months ago    140MB
nginx         latest    0cf1d6af5ca7   3 months ago    161MB
redis         latest    3848bb827e71   3 months ago    140MB
nginx         <none>    fd204fe2f750   4 months ago    161MB
ubuntu        <none>    f794f40ddfff   4 months ago    78.1MB
hello-world   latest    1b44b5a3e06a   11 months ago   10.1kB
redis         <none>    170a1e90f843   2 years ago     138MB

C:\Users\Lenovo>docker run ubuntu /bin/bash
```

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```
C:\Users\Lenovo>docker search ubuntu --filter stars=50
NAME                DESCRIPTION                                     STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx        Nginx, a high-performance reverse proxy & we…   141
ubuntu/kafka        Apache Kafka, a distributed event streaming …   62
ubuntu/apache2      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus   Prometheus is a systems and service monitori…   80
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/mysql        MySQL open source fast, stable, multi-thread…   74

C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\Lenovo>docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
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ubuntu/apache2            Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/postgres           PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   12
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   7
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/alertmanager       Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
ubuntu/loki               Grafana Loki, a log aggregation system like …   2
ubuntu/node               Ubuntu-based Node.js image for server-side a…   5
```

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Conclusion

The objective of containerizing an application using Docker was successfully achieved. Docker provided an isolated and consistent runtime environment by packaging the application and its dependencies into a container. This approach simplified deployment, reduced compatibility issues across different systems, and improved portability and scalability. Overall, Docker proved to be an efficient and reliable platform for application containerization in modern software development.

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